

Product-to-Sum Identities

Write each product as a sum or difference with positive arguments.

1) $\cos 84 \sin 4$

$$\frac{\sin 88 - \sin 80}{2}$$

2) $4 \cos 70 \cos 33$

$$2 \cos 37 + 2 \cos 103$$

3) $-3 \cos 129 \cos 17$

$$\frac{-3 \cos 112 - 3 \cos 146}{2}$$

4) $2 \sin 79 \cos 38$

$$\sin 117 + \sin 41$$

5) $-2 \cos 11B \sin B$

$$-\sin 12B + \sin 10B$$

6) $\sin 5A \sin 7A$

$$\frac{\cos 2A - \cos 12A}{2}$$

7) $\cos 10A \sin A$

$$\frac{\sin 11A - \sin 9A}{2}$$

8) $\cos 4A \cos 8A$

$$\frac{\cos 4A + \cos 12A}{2}$$

Write each sum or difference as a product with positive arguments.

9) $4(\sin 14x - \sin 2x)$

$$8 \cos 8x \sin 6x$$

10) $-5(\cos 91 - \cos 115)$

$$-10 \sin 103 \sin 12$$

11) $\cos 2\theta - \cos 8\theta$

$$2 \sin 5\theta \sin 3\theta$$

12) $3(\cos 15x - \cos 5x)$

$$-6 \sin 10x \sin 5x$$

13) $-5(\cos 3A - \cos 9A)$

$$-10 \sin 6A \sin 3A$$

14) $3(\cos 100 - \cos 248)$

$$6 \sin 174 \sin 74$$

Find the exact value of each expression.

15) $\cos 45 \cos 15$

$$\frac{\sqrt{3} + 1}{4}$$

16) $-3 \cos 195 \sin 45$

$$\frac{3\sqrt{3} + 3}{4}$$

17) $4 \sin 75 \sin 15$

$$1$$

18) $\sin 225 \sin 105$

$$\frac{-1 - \sqrt{3}}{4}$$

19) $\sin 105 \cos 225$

$$\frac{-1 - \sqrt{3}}{4}$$

20) $2 \cos 195 \sin 45$

$$\frac{-\sqrt{3} - 1}{2}$$